Likelihood and the Probability Scale

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS likelihood · impossible *(tē taea)* · unlikely *(tūpono iti)* · even chance · likely *(tūpono nui)*

A Likelihood words ●○○

Remember: Impossible: it cannot happen. Unlikely: it probably will not happen. Even chance: it is just as likely to happen as not. Likely: it probably will happen. Certain: it must happen.

1. Choose the best word for each event: impossible, unlikely, even chance, likely or certain.

a) Getting a head when a fair coin is tossed _____

b) Rolling a 6 on a fair dice _____

c) Rolling a number less than 7 on a fair dice _____

d) Rolling a number greater than 1 on a fair dice _____

e) Rolling a 9 on a fair dice _____

f) Picking a red counter from a bag that holds 1 red and 9 blue counters _____

g) Picking a blue counter from the same bag _____

h) Rolling an even number on a fair dice _____

B Ordering by likelihood ●●○

2. Put the three events in order, from least likely to most likely. Write the letters.

a) A: Rolling a number less than 6 on a fair dice. B: Rolling a 6 on a fair dice. C: Getting a head on a fair coin.

b) A: Picking a red counter from a bag of 5 red counters. B: Picking a red counter from a bag of 5 blue counters. C: Picking a red counter from a bag of 2 red and 3 blue counters.

c) A: Rolling an even number on a fair dice. B: Rolling a 2 on a fair dice. C: Rolling a number greater than 2 on a fair dice.

C The probability scale ●●○

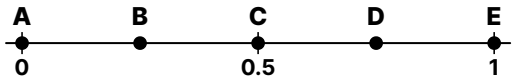
Remember: A probability is a number from 0 to 1. Impossible is 0, an even chance is $\frac{1}{2}$ and certain is 1.

3. Write the probability, as a number, of an event that is:

a) impossible _____ **b)** certain _____ **c)** an even chance _____

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4. Which letter on the scale shows the probability of each event?



a) Getting a tail when a fair coin is tossed _____

b) Picking a red counter from a bag that holds 1 red and 3 blue counters _____

c) Picking a blue counter from the same bag _____

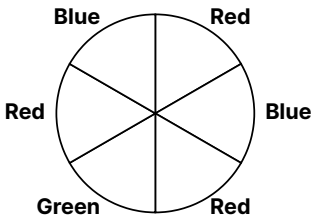
d) Rolling a 7 on a fair dice _____

e) Rolling a number less than 7 on a fair dice _____

D Sample space ●●●

> **Remember:** The sample space is the list of all the possible outcomes. For a coin it is: heads, tails. Be systematic, so that you do not miss any out.

5. This spinner has 6 equal parts.



a) Write the sample space (every colour it can land on).

b) Which colour is it most likely to land on?

c) Which colour is it least likely to land on?

6. List all the possible outcomes. Use the letters and numbers given.

a) A fair dice is rolled.

b) A coin is tossed twice (H, T).

c) A coin is tossed (H, T) and a spinner numbered 1, 2, 3 is spun.

7. How many outcomes are there?

a) A coin is tossed

b) A dice is rolled

c) A coin is tossed and a dice is rolled

d) 2 shirts and 3 pairs of shorts

★ Extension

8. You choose a sandwich (cheese C, ham H, egg E) and a drink (juice J, water W). List all the possible lunches.

9. Give an example of an event that is certain, and an example of an event that is impossible.

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Likelihood and the Probability Scale

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS likelihood · impossible *(tē taea)* · unlikely *(tūpono iti)* · even chance · likely *(tūpono nui)*

A Likelihood words ●○○

Remember: Impossible: it cannot happen. Unlikely: it probably will not happen. Even chance: it is just as likely to happen as not. Likely: it probably will happen. Certain: it must happen.

1. Choose the best word for each event: impossible, unlikely, even chance, likely or certain.

a) Getting a tail when a fair coin is tossed _____

b) Rolling a 1 on a fair dice _____

c) Rolling a number greater than 0 on a fair dice _____

d) Rolling a number less than 6 on a fair dice _____

e) Rolling a 0 on a fair dice _____

f) Picking a blue counter from a bag that holds 9 red and 1 blue counter _____

g) Picking a red counter from the same bag _____

h) Rolling an odd number on a fair dice _____

B Ordering by likelihood ●●○

2. Put the three events in order, from least likely to most likely. Write the letters.

a) A: Getting a tail on a fair coin. B: Rolling a number greater than 1 on a fair dice. C: Rolling a 3 on a fair dice.

b) A: Picking a blue counter from a bag of 4 red counters. B: Picking a blue counter from a bag of 3 blue counters and 1 red counter. C: Picking a blue counter from a bag of 4 blue counters.

c) A: Rolling a number less than 3 on a fair dice. B: Rolling an odd number on a fair dice. C: Rolling a 5 on a fair dice.

C The probability scale ●●○

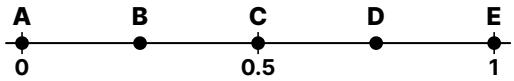
Remember: A probability is a number from 0 to 1. Impossible is 0, an even chance is $\frac{1}{2}$ and certain is 1.

3. Write the probability, as a number, of an event that is:

a) impossible _____ **b)** certain _____ **c)** an even chance _____

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4. Which letter on the scale shows the probability of each event?



a) Getting a head when a fair coin is tossed _____

b) Picking a green counter from a bag that holds 3 green and 1 red counter _____

c) Picking a red counter from the same bag _____

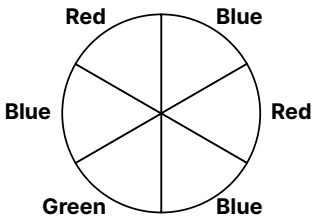
d) Rolling an 8 on a fair dice _____

e) Rolling a number greater than 0 on a fair dice _____

D Sample space ●●●

> **Remember:** The sample space is the list of all the possible outcomes. For a coin it is: heads, tails. Be systematic, so that you do not miss any out.

5. This spinner has 6 equal parts.



a) Write the sample space (every colour it can land on).

b) Which colour is it most likely to land on?

c) Which colour is it least likely to land on?

6. List all the possible outcomes. Use the letters and numbers given.

a) A spinner numbered 1 to 5 is spun.

b) A coin is tossed twice (H, T).

c) A coin is tossed (H, T) and a spinner numbered 1, 2, 3, 4 is spun.

7. How many outcomes are there?

a) A coin is tossed

b) A dice is rolled

c) A coin is tossed and a dice is rolled

d) 3 shirts and 2 pairs of shorts

★ Extension

8. You choose an ice cream (vanilla V, chocolate C, mint M) and a topping (nuts N, sauce S). List all the possible choices.

9. Give an example of an event that is certain, and an example of an event that is impossible.

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